1. Introduction

Cardiovascular diseases are among the leading causes of death worldwide. Early detection of heart rhythm abnormalities (arrhythmias) is critical for preventing severe cardiac events such as heart attacks and strokes. Electrocardiogram (ECG) signals provide vital information about the electrical activity of the heart and are commonly used for arrhythmia detection.

Traditional ECG monitoring systems rely on software-based signal processing methods that require significant computational resources and power. These methods are not suitable for portable or wearable medical devices where low power consumption and real-time processing are essential.

Application-Specific Integrated Circuits (ASICs) offer an efficient solution for implementing dedicated hardware accelerators for biomedical signal processing. By designing specialized hardware architectures, ECG signal processing and classification can be performed with lower latency and power consumption.

Approximate computing is an emerging design technique that intentionally introduces small computational inaccuracies to significantly reduce power consumption, circuit area, and delay. In biomedical signal processing, slight approximations are often acceptable because biological signals inherently contain noise and variability.

This project aims to design and implement an ASIC-based real-time ECG arrhythmia classifier using approximate computing techniques. The proposed system will process ECG signals, extract important features such as RR intervals, and classify heartbeats as normal or abnormal using hardware-efficient algorithms.

2. Background Study

Electrocardiogram signal analysis has been extensively studied for automatic arrhythmia detection. Several algorithms have been developed for ECG signal preprocessing, QRS complex detection, feature extraction, and classification.

One of the most widely used techniques for QRS detection is the Pan–Tompkins algorithm, which includes filtering, differentiation, squaring, and thresholding operations to accurately detect R-peaks in ECG signals.

Machine learning techniques such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Convolutional Neural Networks (CNN) have also been applied for ECG classification. However, these approaches often require high computational power and memory resources.

To address these limitations, hardware-based ECG processing architectures have been proposed using FPGA and ASIC platforms. Hardware implementations offer advantages such as reduced power consumption, faster processing, and improved real-time performance.

Approximate computing techniques have recently gained attention for energy-efficient hardware design. Approximate adders, multipliers, and arithmetic units can significantly

reduce power and area while maintaining acceptable accuracy levels for error-tolerant applications like signal processing and machine learning.

Recent research shows that integrating approximate arithmetic units into biomedical hardware accelerators can achieve substantial improvements in power efficiency while maintaining reliable classification performance.

3. Objectives and Problem Formulation

The primary objective of this project is to design and implement a low-power ASIC chip that can automatically detect abnormal heart rhythms (arrhythmias) from ECG signals in real time, using approximate computing techniques to reduce power consumption.

Specific Objectives

- To study how ECG signals work, what normal and abnormal heartbeats look like, and how different arrhythmias such as APC, PVC, Ventricular Tachycardia, Atrial Fibrillation, and Paced Beats can be identified.
- To design an ECG signal preprocessing block that removes noise and extracts useful features like QRS shape and RR interval.
- To design a deeper neural network classifier that can accurately distinguish between different types of arrhythmias, including clinically distinct conditions that existing designs incorrectly merge into one class.
- To apply approximate computing techniques to the classifier's arithmetic units to reduce power and hardware area without significantly affecting accuracy.
- To implement the complete system architecture using Verilog HDL at the RTL level.
- To perform functional simulation to verify the correctness of the RTL design.
- To synthesize the design using standard cell libraries.
- To perform physical design including floor planning, placement, clock tree synthesis, and routing.
- To perform physical verification using DRC and LVS checks.
- To generate the final GDSII layout file of the ASIC.

Problem Statement

The heart beats in a regular rhythm controlled by electrical signals. When this rhythm becomes abnormal — beating too fast, too slow, or irregularly — it is called an arrhythmia. Some arrhythmias are immediately life-threatening (like Ventricular Tachycardia), while others are early warning signs of worsening heart health (like Extreme Bradycardia). Since arrhythmias can occur suddenly and at any time, continuous real-time monitoring is essential to save lives.

An ECG records the heart's electrical activity as a wave pattern. By analyzing the shape of the QRS complex (the main heartbeat signal) and the time gap between beats (RR interval), it is possible to automatically identify what type of arrhythmia is occurring.

Currently, ECG classification systems running on software or general-purpose processors consume too much power to be used in small wearable or implantable medical devices. Dedicated ASIC chips have been developed to solve this, but they still have major problems:

- Their neural network is too shallow — it can only identify 4 types of QRS shapes and even merges two clinically different conditions (PVC Type II and Paced Beat) into one class, which is a serious medical error
- 80% of the chip's power is wasted just on memory storage
- The noise filtering is too basic and cannot handle real-world motion artifacts
- The chip still needs external components (AFE and ADC), so it cannot work alone

Therefore, there is a need for a dedicated low-power ASIC that uses approximate computing — a technique where small, controlled errors are allowed in mathematical calculations to save significant power and area — to support a deeper, more accurate classifier that correctly identifies all arrhythmia types and is suitable for real wearable medical devices.

4. Proposed Methodology

The proposed system will follow a hardware-based signal processing approach for ECG arrhythmia classification.

The ECG signal processing flow includes the following stages:

1. ECG Signal Acquisition
2. Signal Preprocessing
3. QRS Complex Detection
4. Feature Extraction
5. Arrhythmia Classification

Implementation Steps

1. Study ECG datasets such as the MIT-BIH Arrhythmia Database.
2. Design ECG preprocessing and feature extraction algorithms.
3. Implement the system architecture using Verilog HDL (RTL design).
4. Perform functional simulation to verify the correctness of the RTL design.
5. Perform logic synthesis using ASIC standard cell libraries.
6. Carry out floorplanning to define the chip layout and power structure.
7. Perform placement of standard cells within the chip area.
8. Apply clock tree synthesis (CTS) to distribute the clock signal efficiently.
9. Perform routing to connect all components using metal interconnects.
10. Conduct physical verification including DRC and LVS checks.

11. Generate the final GDSII layout file.

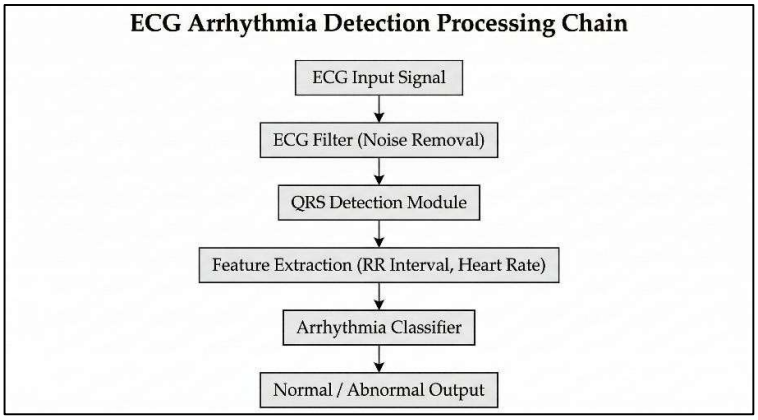

Figure 4.1: ECG Arrhythmia Detection Processing Chain

Figure 4.1 illustrates the processing chain of the proposed ECG arrhythmia detection system. The ECG signal is first filtered to remove noise, followed by QRS detection and feature extraction. The extracted features are then processed by the arrhythmia classifier to determine whether the heartbeat is normal or abnormal.

5. Software & Hardware Requirements

- Cadence Xcelium – Used as the simulation tool for verifying the RTL design and functional behavior of the ECG arrhythmia detection system.
- Cadence Genus – Used for logic synthesis, converting the RTL design written in Verilog into a gate-level netlist using standard cell libraries.
- Cadence Innovus – Used for physical design, including floorplanning, placement, clock tree synthesis (CTS), and routing of the ASIC design.
- Cadence Virtuoso – Used for schematic design, circuit simulation, layout design, physical verification (DRC/LVS), and generation of the final GDSII file.
- MATLAB / Python – Used for ECG signal analysis and preprocessing of ECG datasets, such as the MIT-BIH Arrhythmia Database, to understand signal characteristics and extract relevant features.

Hardware Requirements

- Personal Computer / Laptop for accessing the remote Cadence server.
- Stable Internet Connection for VPN-based remote access to the Cadence environment.

6. References

[1] J. Pan and W. J. Tompkins,
"A Real-Time QRS Detection Algorithm," *IEEE Transactions on Biomedical Engineering*, vol. 32, no. 3, pp. 230–236, 1985.

[2] H. Lee, Y. Kim, and H. Yoo,

"A 58-nW ECG ASIC With Motion-Tolerant Heartbeat Timing Extraction for Wearable Cardiac Monitoring Devices," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 9, no. 5, pp. 635–644, 2015.

[3] S. M. Bhandari, J. S. Park, and H. Yoo,
"A 9.6-μW ECG Processor With QRS Detection and Arrhythmia Classification for Wearable Cardiac Monitoring Devices," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 14, no. 3, pp. 623–635, 2020.

[4] H. Lee, Y. Kim, and S. Yoo,
"A Low-Power ECG Signal Processing Processor With Adaptive Thresholding for RealTime Arrhythmia Detection," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 15, no. 4, pp. 789–800, 2021.

[5] Y. Kim, J. Park, and H. Yoo,
"A Wearable ECG Monitoring System With Real-Time Arrhythmia Detection Using LowPower Hardware Architecture," *IEEE Access*, vol. 9, pp. 115234–115245, 2021.

[6] S. Kiranyaz, T. Ince, and M. Gabbouj,
"Real-Time Patient-Specific ECG Classification by 1-D Convolutional Neural Networks," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 3, pp. 664–675, 2016.

[7] Z. Zhang, J. Dong, X. Luo, K. S. Choi, and X. Wu,
"Heartbeat Classification Using Deep Neural Networks," *IEEE Access*, vol. 7, pp. 149882–149889, 2019.

[8] U. R. Acharya, H. Fujita, V. K. Sudarshan, S. L. Oh, and M. Adam,
"Application of Deep Convolutional Neural Network for Automated Detection of Myocardial Infarction Using ECG Signals," *IEEE Access*, vol. 5, pp. 14100–14110, 2017.

[9] G. Moody and R. G. Mark,
"The MIT-BIH Arrhythmia Database on CD-ROM and Software for Use With It," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.

7. Project Base Paper

"A 746 nW ECG Processor ASIC Based on Ternary Neural Network"

Authors: Syed Muhammad Abubakar, Yue Yin, Songyao Tan, Hanjun Jiang, Zhihua Wang

Publisher: IEEE Transactions on Biomedical Circuits and Systems, Vol. 16, No. 4 Date:

August 2022